



Genesis and metallogenetic setting of the polymetallic barite-sulphide deposit, Bobija, Western Serbia

Marcel Maurer¹ · Dejan Prelević^{1,2} · Regina Mertz-Kraus¹ · Aleksandar Pačevski² · Bojan Kostić² · Jovana Malbašić²

Received: 6 December 2018 / Accepted: 13 May 2019 / Published online: 25 May 2019
© Geologische Vereinigung e.V. (GV) 2019

Abstract

The polymetallic Bobija ore deposit in Western Serbia is located at the southwestern edge of the Jadar Block, a distal part of the passive continental margin of Adria. The sulphide mineralization consists of fine-grained pyrite, sphalerite, galena and tetrahedrite, and is associated with abundant barite. The deposit is stratabound and hosted by a Triassic sedimentary sequence. A feeder zone is present in the immediate footwall of the ore mineralization. The volcano-sedimentary succession in the uppermost strata comprises coherent lavas and volcanoclastic sediments. They were investigated for the bulk rock composition and the zircons were subjected to U–Pb dating. Analysed zircons deliver an upper age limit of 243.1 ± 1.3 Ma. Sphalerite from the sulphide mineralization was investigated for its trace element content and pyrite and barite for their sulphur isotope composition. Sphalerite geothermometry indicates ore-bearing fluid temperatures of 152–179 °C. A negative $\delta^{34}\text{S}$ value of -6.3‰ in pyrite indicates a sulphur source from a partially reduced reservoir. The new data confirm a previously inferred Triassic age for the Bobija deposit and underline the importance of the prevalent rift regime for ore formation. We propose that the sediment-hosted Bobija deposit should be classified as a clastic-dominated, rift-related Pb–Zn deposit. Given the tectonic setting and temperature constraints, the hydrothermal processes are comparable to deposits classically referred to as sedimentary exhalative (SEDEX). This is in a good agreement with similar deposits occurring in Central Bosnia. These similarities reveal links between ore deposit formation and geodynamic evolution in the Balkan Peninsula, underlining the common geological evolution of the more external zones of the Dinarides and the Jadar Block.

Keywords Trace elements in sphalerite · Sulphur isotopes · Zircon dating · SEDEX · SHMS · Clastic-dominated Pb–Zn · Sulphides · Ore geology

Introduction

The ore deposits of the Balkan Peninsula are part of the Alpine-Balkan-Carpathian-Dinaride (ABCD) belt and constitute one of the world's oldest mining areas (Heinrich and Neubauer 2002). Episodes of ore mineralization are closely

linked to the complex geodynamic history of the region, resulting in variability of deposit types that strongly correlates with geologic epochs (e.g. Janković 1997; Palinkaš et al. 2008a). Examples are the Permian–Triassic rift-related sedimentary exhalative (SEDEX) and Mississippi Valley-type (MVT) deposits in the Dinarides (Palinkaš et al. 2008a, 2016), Jurassic ophiolite-related chromite, asbestos, talc and magnesite deposits (Janković 1997; Palinkaš et al. 2008a), Cretaceous porphyry Cu, Mo and Au deposits in relation to subduction of oceanic lithosphere (Berza et al. 1998; Strashimirov and Popov 2000), and the Tertiary post-collisional hydrothermal Pb, Zn, Ag and Sb deposits coupled with intrusive and extrusive magmatism (Janković 1997; Palinkaš et al. 2008a; Radosavljević et al. 2013a). Hence, the age of a deposit in the Balkan Peninsula is a strong indicator of its geotectonic setting, dictating possible ore sources and the transport and deposition mechanisms.

Electronic supplementary material The online version of this article (<https://doi.org/10.1007/s00531-019-01732-8>) contains supplementary material, which is available to authorized users.

✉ Marcel Maurer
marcelmaurer@live.com

¹ Institute for Geosciences, Johannes Gutenberg University, 55099 Mainz, Germany

² Faculty of Mining and Geology, University of Belgrade, Belgrade 11000, Serbia

The polymetallic barite-sulphide deposit of Bobija (Fig. 1) is located within the Jadar Block, which represents one of several tectonic units derived from parts of the ancient Adria margin (Karamata 2000; Schmid et al. 2008; Robertson et al. 2009; Cvetković et al. 2016). In terms of mineralogy and origin, the Bobija deposit is distinct from its neighbouring Tertiary ore deposits associated with the Oligocene Boranja intrusion (Radosavljević et al. 2013a, b, 2016). Based on the volcano-sedimentary succession, it was classified as a volcano-sedimentary deposit related to Triassic rift magmatism (Janković 1990). More recent investigations focused on the mineralogical characterisation of sulphides and sulphosalts (Radosavljević et al. 2013b, 2016) and processing of the sulphide and barite ore (Vrčar et al. 2003). However, none of the previous studies constrained the radiogenic age of the associated volcanism and the conditions of ore formation, as well as evaluated the thermochemical environment that was a prerequisite for the mineralization processes.

In this paper, we reassess previous studies and produce a new set of geochemical and mineralogical data in an attempt to constrain more thoroughly the timing of magmatism, its relationship to the ore formation processes and physicochemical conditions of ore formation. We produce an internally consistent dataset of good quality, which enables inter-province comparison with similar deposits in the Balkan Peninsula. We use trace element concentrations in sphalerite obtained by LA-ICPMS, to constrain the ore-bearing fluid temperatures. Using sulphur isotope composition of pyrite and barite, we gain information about the sulphur source. Parallely, we study the volcano-sedimentary succession including volcanoclastic sediments and coherent basaltic lava associated with the mineralization in Bobija, to

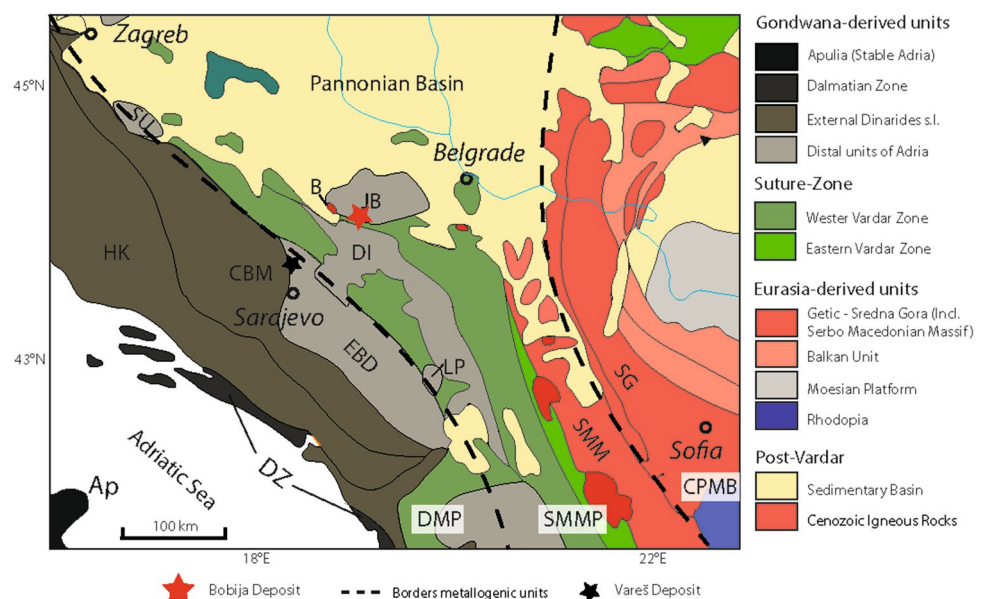
determine their age and geotectonic setting. All these results are discussed within the geotectonic framework of the Balkan Peninsula. Finally, a model for the metallogenes of the Bobija deposit is proposed.

Geological setting

The regional geology of the Balkan Peninsula dominantly originates from the Mesozoic events at the north-eastern edge of Adria, namely continental rifting and subsequent seafloor spreading between the Adriatic microplate and Eurasia, followed by plate convergence with ophiolite obduction and eventually collision in an orogenic process (e.g. Ansorge et al. 1993). This is directly linked to the opening and closure of a branch of the Mesozoic Tethys located between Eurasia and Adria, called ‘Vardar Tethys’ (Cvetković et al. 2016) or ‘Neotethys’ (Schmid et al. 2008). Geological units pre-dating the opening of the Vardar Tethys are represented by remnants of Adria and its distal parts exposed within the suture zone itself. The onset of ocean floor spreading was preceded by a phase of rifting in the northern part of Adria from the late Permian time onwards (e.g. Palinkaš et al. 2008a; Robertson et al. 2009). The divergence between Adria and Moesia (Europe) led to subsidence and formation of a number of intracontinental rift-basins. Hereby, two rift-phases are distinguished: a Permian early intra-continental stage, and a Triassic advanced rifting stage (Palinkaš et al. 2008a). The latter is closely associated with occurrences of Triassic basalts, which intruded unconsolidated Triassic sediments (Palinkaš et al. 2008b; Kiss et al. 2012, 2016a).

The Bobija deposit is located in Western Serbia, at the southwestern edge of the Adria-derived Jadar Block (Fig. 1).

Fig. 1 Tectonic Map of the Northern Balkan Peninsula, modified after Dimitrijević (1992). *Ap* Apulia; *B* Boranja Intrusion; *CBM* Central Bosnian Mountains Unit; *DI* Drina-Ivanjica Unit; *DZ* Dalmatian Zone; *EBD* East-Bosnian Dinaric Unit; *HK* High Karst nappe; *JB* Jadar Block; *LP* Lim Paleozoic; *SG* Sredna-Gora; *SMM* Serbo-Macedonian Massif; *SU* Sana-Una Unit. Metallogenic units of Serbia: *DMP* Dinaric Metallogenic Province; *SMMP* Serbian-Macedonian Metallogenic Province; *CPMB* Carpatho-Balkan Metallogenic Province. After Jelenković et al. (2008). Abbreviations of lithological units are taken from Cvetković et al. (2016)



The pre-Permian succession of the Jadar Block is built up of Devonian to Upper Carboniferous siliciclastic sediments and shallow-marine carbonates, followed by fusulinid limestones and siltstones up to earliest Permian time. Sedimentation continued in the mid-Permian with deposition of clastic sediments, passing into gypsum, dolomite and limestone. Triassic lithologies are present as shallow-water limestones and basaltic—intermediate volcanic rocks (Protić et al. 2000). The Jadar Block is generally stratigraphically similar to the Paleozoic Sana-Una and the Bükk Unit (Filipović et al. 2003), as well as to the South-Eastern Alps (Haas et al. 1995). Its exact geodynamic history is still a matter of an ongoing debate. Schmid et al. (2008) interpret the Jadar Block to structurally underlie remnants of the Neotethys Ocean that were obducted onto the Adria margin. In contrast, Karamata (2000) considers the Jadar Block a terrane that was detached from the north-western margin of Adria by an additional oceanic tract.

Two types of mineralization are generated during the Triassic advanced rifting stage at the north-eastern edge of Adria: Mississippi Valley-type (MVT) and sedimentary-exhalative (SEDEX) deposits (Palinkaš et al. 2008a). The MVT (or ‘Bleiberg-Mežica’) type deposits are situated in the Mid-Triassic carbonate platform formations of the External Dinarides. They are characterised by simple Pb–Zn parageneses without Cu and low formation temperatures (100–200 °C). The SEDEX deposits are interpreted to be the result of seafloor hydrothermal exhalations, occasionally associated with prominent feeder zones. The parageneses include pyrite, cinnabar, base metal sulphides and sulphosalts, barite, siderite, hematite and Mn-oxides (Palinkaš et al. 2008a).

Sampling and field relations

The Bobija deposit is located at the eastern slopes of the Sokolske Mountains, about 14 km NE from the city of Ljubovija (Fig. 1). It is a polymetallic deposit, mostly composed of barite and paragenesis of barite-pyrite-sphalerite-galena (Radosavljević et al. 2013b, 2016). Janković (1987) linked the origin of the deposit with the Mid-Triassic rift magmatism. The Bobija mine is almost exhausted and closed, and only recently an exploration company has started intense exploration in this area.

Figure 2 shows the cross-section of a mineralized outcrop not larger than 35 × 5 m that remained after previous exploitation and Fig. 3 displays a synthetic stratigraphic lithological profile of the Bobija deposit, including the related basement and cover units comprised of volcano-sedimentary succession. A list and description of investigated samples are given in Online Resource 1. An overview of sampling positions is given in Online Resource 2. In the course of this study, we took three massive sulphide samples (16B06; 16B07; 16B09_2), one barite sample (16B12), two sediment samples (16B14; 16B15), two volcanoclastic sediments (volcanoclastic sandstone 16B10; volcanoclastic mudstone 16B11) and one sample of coherent lavas (16B13; see Online Resource 1 and Fig. 2). It is confirmed by the geologists who are performing exploration-drilling around the Bobija deposit, that our samples are compositionally similar to the material retrieved from the boreholes. Unfortunately, samples from the feeder zone were not available for investigation.

The sulphides at Bobija display a massive texture and mainly consist of pyrite and smaller amounts of galena, sphalerite and tetrahedrite. Intercalated within the massive sulphides are irregularly distributed lenses of barite. The barite contains minor amounts of disseminated sulphides. The sulphide and barite ore appear brecciated with the matrix of

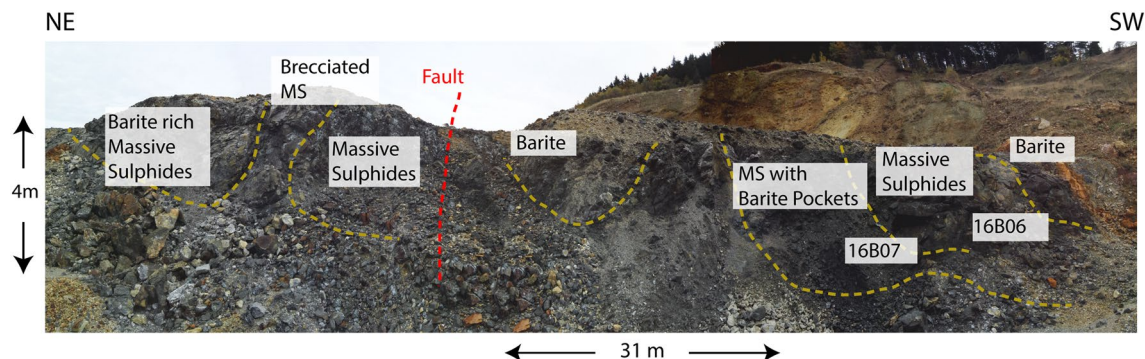


Fig. 2 Cross section of the sampled massive sulphide (MS) outcrop. Marked are sample location and the fault cutting through the outcrop

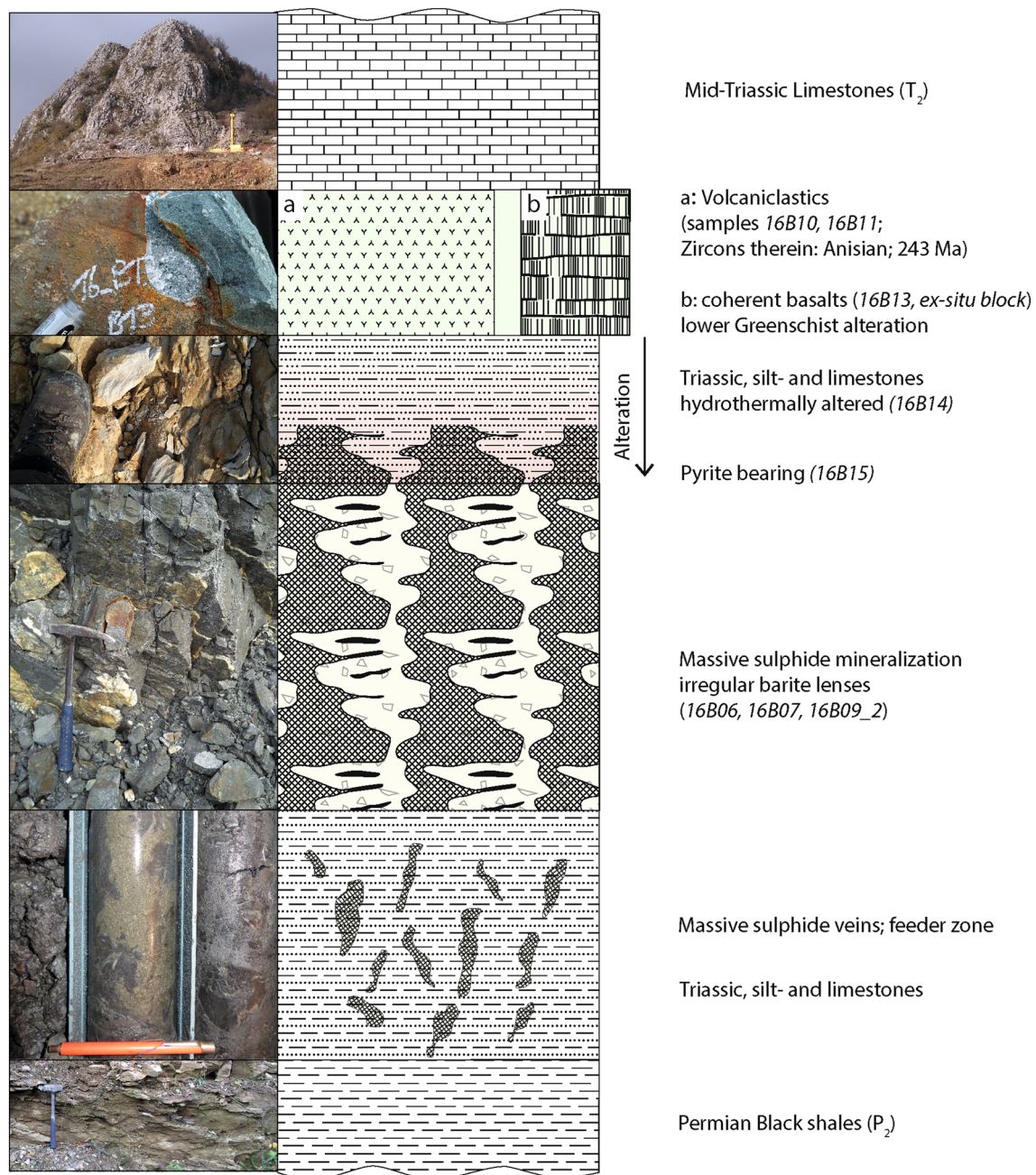


Fig. 3 Schematic lithological profile of the Bobija deposit. Samples are given at the units where they have been collected from. The interbedded shale and limestone formation in the footwall is not exposed at the mine site. Formation thicknesses are not true to scale, their proportions relative to each other are estimated from field observations.

the same composition as the clasts and are further crosscut by a shear zone (Fig. 3). The lowermost strata below the outcropping massive sulphides are composed of Triassic sand/siltstones and limestones (Fig. 3). The exact age is not constrained. Within the limestone/siltstone succession, sulphide veins are present (personal communication, exploration geologists). Below, Permian black shales (P_2) have been drilled.

Note that the volcaniclastic sediments and the coherent basalt sample are not directly associated in the field, but similarities in geochemistry argue towards a similar origin (see “[The volcano-sedimentary succession](#)” section). The exact age of the silt- and limestone formation is not constrained

Stratigraphically above and conformably to the main massive sulphide outcrop, hydrothermally altered sandstones and siltstones appear, with the most proximal ones containing abundant pyrite (sample 16B15). Atop these sediments, volcaniclastic sediments (samples 16B10 and 16B11) occur. We assume that both, altered sandstones/siltstones, and volcaniclastic sediments lie conformably and are of similar age.

Basalt (sample 16B13) was collected from a block about 100 m SE of the volcanoclastic sediments. This block was not in situ and hence its stratigraphic position cannot be clearly established from the field. As we discuss below, based on its bulk rock chemistry (“[The volcano-sedimentary succession](#)” section) we expect that the lava represents part of the same volcano-sedimentary succession. On the top of this succession Mid-Triassic limestones (T_2) outcrop.

Analytical methods

In the following section, a brief overview of the sample preparation and applied methods is given. A more detailed description together with the particular instrumental settings for each method is enclosed as Online Resource 3.

All samples were cut into rectangular slabs of 2 cm × 3 cm. These were subsequently placed on a cover-slip and ground down to a thickness of 30 µm. The massive sulphide samples were cut into 3 cm × 2 cm × 1 cm cuboids. These were placed in epoxy resin and were subsequently polished. Bulk rock trace element analyses (samples 16B10, 16B11, 16B13), as well as zircon U–Pb age determination (16B10), were performed using the LA-ICPMS at the Department of Geosciences, the University of Mainz with an ESI NWR193 ArF 193 nm excimer laser system equipped with a TwoVol² ablation cell coupled to an Agilent 7500ce quadrupole ICPMS. The bulk rock trace element analyses were performed on glass beads gained by utilizing an iridium strip heater according to Stoll et al. (2008). Bulk rock major element data was acquired using Philips MagiXPRO spectrometer on fused discs (XRF) at the Department of Geosciences of the University of Mainz.

Major and trace element compositions of sphalerites (samples 16B06, 16B07, 16B09_2) were investigated using a JEOL JXA-8200 electron microprobe with a Wavelength Dispersive Spectrometry (WDS) analytical system at the Johannes Gutenberg-University, Mainz and utilizing the same LA-ICPMS as for the bulk-rock trace element analyses above. In addition to sphalerite, trace element data of galena, pyrite and tetrahedrite were acquired. Although not discussed in this study, they are of good quality and might be of future interest for studies regarding trace elements in sulphides. They are enclosed as a part of Online Resource 7. The complete datasets of performed analyses are enclosed in the Electronic Supplementary Material (ESM) and only representative data are given in the paper.

For sulphur isotope investigation, pyrite (Py2 type see below) and barite powders were prepared using an ESI New Wave micro mill. Simple hand-picking was not feasible due to small grain sizes. Since the amount of retrieved pyrite powder was insufficient, we mixed the pyrite powders gained from each of the massive sulphide samples (16B06, 16B07,

16B09_2). Barite from samples 16B06, 16B07 and 16B12 (pure barite vein) was retrieved and analysed separately. The sulphur isotope measurements were performed at the Scientific and Technological Center at the University of Barcelona in an isotope ratio mass spectrometer (IRMS) Delta Plus XP (ThermoFisher).

Results

In the following, a brief description of the samples is given, and geochemical data are presented. A more detailed petrographic description is given as part of Online Resource 1.

The volcano-sedimentary succession

The lithologies from the volcano-sedimentary succession comprise volcanoclastic sediments (16B10, 16B11) and coherent lava (16B13; not in situ). Since the volcanoclastic sediments and the coherent lava are not directly associated in the field (displayed in Online Resource 2; sample locations), it was unclear whether they represent a part of the same volcano-sedimentary succession and whether conclusions for the coherent basalt are automatically valid for the volcanoclastic sediments and vice versa. This is especially important for the U–Pb age determination of the zircons retrieved from the volcanoclastic rocks. As we demonstrate below, a notable geochemical resemblance is apparent, which supports their similar magmatic origin.

Stratigraphically, the volcanoclastic sediments overlay the fluid altered sediments (Fig. 3). They have potentially been mixed with associated unconsolidated strata and hence could comprise non-volcanic clasts. Such mixing is not observed in any of the samples but might be obscured by the strong alteration and hence cannot be ruled out. All samples from the volcano-sedimentary succession, including volcanoclastic sediments and coherent lava, are extensively altered with the presence of similar secondary mineral paragenesis, hindering comprehensive microprobe analysis of the major mineral phases: feldspar is pseudomorphed by sericite and amphibole, whereas pyroxene grains are replaced by chlorite. Such strong alteration is also a general feature of Triassic magmatic rocks in the Central Dinarides (Bosnia-Herzegovina) that are oftentimes referred to as spilites and keratophyres (e.g. Pamić 1984; Trubelja et al. 2004).

Bulk rock data of the three samples is given in Online Resource 4 and is displayed in Fig. 4. Due to the high grade of alteration, only fluid immobile elements are suitable for classification. Geochemically, their composition corresponds to andesite (16B10) and alkaline basalt (16B11 and 16B13) (Winchester and Floyd 1977; Fig. 4a). Figure 4b discriminates the volcanic samples based on immobile elements after Meschede (1986). The Zr/4–Y–2*Nb diagram infers

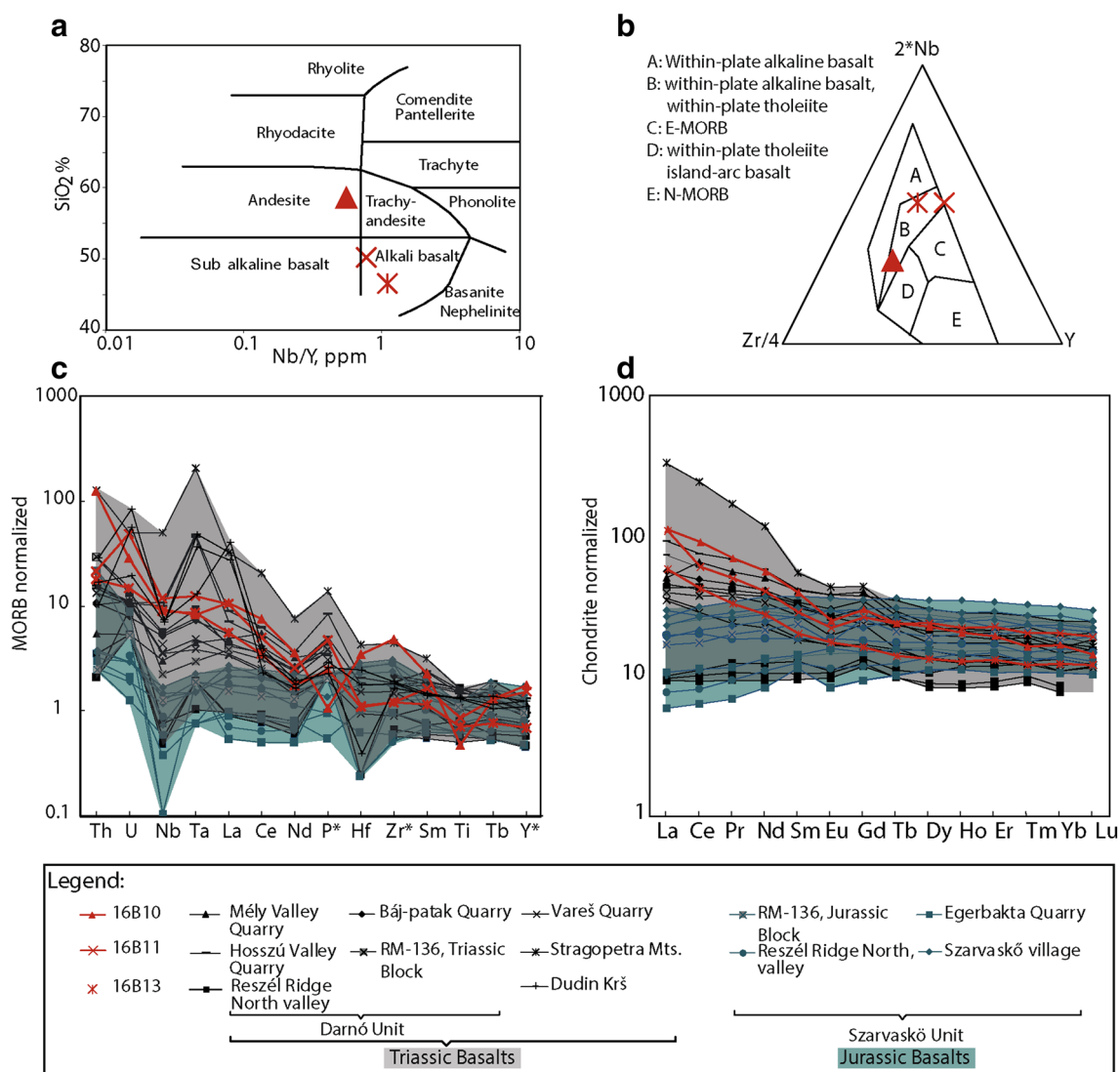


Fig. 4 **a** Classification of volcanic samples 16B10, 16B11 and 16B13 after Winchester and Floyd (1977). **b** Discrimination diagram to distinguish between different types of MORB and continental tholeiites after Meschede (1986). **c**, **d** MORB normalized (Pearce and Parkinson 1993) incompatible and fluid immobile elements (**c**) and chondrite normalized (McDonough and Sun 1995) REEs (**d**) from bulk rock analyses of volcanic samples in comparison to Triassic (grey)

and Jurassic (blue) basalts from the Dinarides (Bükk Unit in NE Hungary; Kalnik Mts. in Croatia; Vareš area in Bosnia Herzegovina) and Hellenides (Stragopetra Mts. in Greece). Respective basalt data is taken from Kiss et al. (2012) and Kiss et al. (2016a). Data for the Triassic metabasalts of Dudin Krš are taken from Memović et al. (2003). Elements marked with * are taken from XRF analysis

a within-plate origin for all samples from the volcano-sedimentary succession.

Incompatible trace elements diagrams (Fig. 4c, d) show a compilation of trace element and REE patterns of Triassic and Jurassic basalts from the Dinarides and Hellenides (Triassic: Memović et al. 2003; Jurassic and Triassic: Kiss et al. 2012, 2016a) in comparison to the samples of the lithologies of the volcano-sedimentary succession found at Bobija. Additional data on Triassic rifting magmatism can be found in Karamata et al. (2000) and Knežević and Cvetković (2000). Figure 4c and d display incompatible and fluid

immobile element patterns. Enrichment in incompatible elements and, especially, similar fractionation of less- relative to more-compatible elements (e.g. light REE vs. heavy REE) in the volcanoclastic sediments and coherent lava indicates their cogenetic origin and generally low degrees of partial melting, within the array of Triassic basalts from the wider region. LREEs of the Bobija lava are elevated in comparison to HREEs, similar to other Triassic basalts especially from the Darnó Unit within the Bükk region in NE Hungary, as well as basalts from the Vareš region. HREE trends resemble

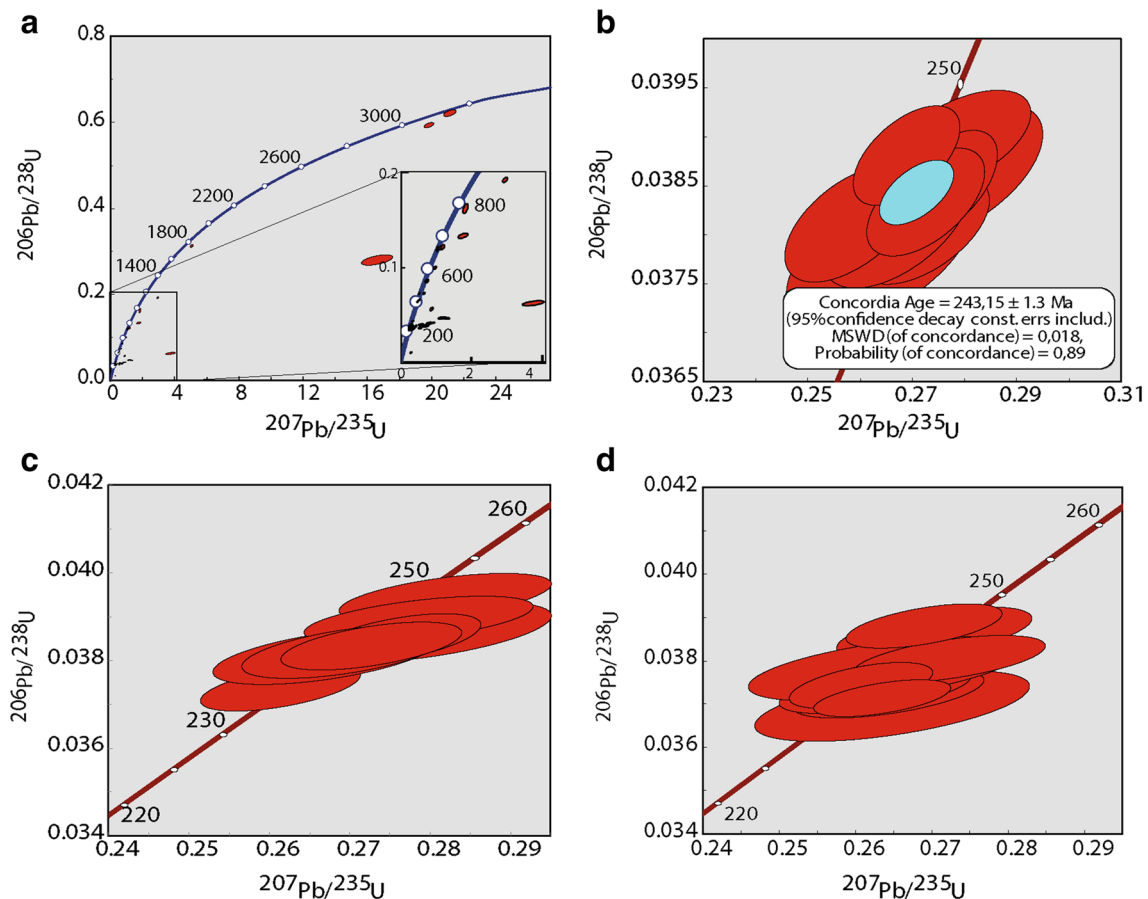


Fig. 5 Resulting Wetherill concordia diagram from U–Pb zircon analysis. **a** Visualizes all obtained data. **b** Calculated concordia age for sample 16B10. **c** Assemblage of concordant data from zircon cores

d: Assemblage of concordant data from zircon rims. Diagrams have been calculated with Isoplot 4.15 (Ludwig 2012). Data point error eclipses are 2σ

trends at Vareš, but LREE (La, Ce, Pr, Nd) are about a factor of 1.5 higher.

The results of U–Pb zircon analysis of the volcanoclastic sediment sample 16B10 are displayed in Fig. 5. Corresponding data and CL images are given as Online Resource 5 and 6. Concordant data ranges from Triassic to Archean in age. Four populations can be distinguished: Mid-Archean, Neo-proterozoic, Early Devonian and Mid-Triassic. This indicates the involvement of continental crust during the magmatic evolution and the eruption event. The focus of this study is on the most recent magmatic activity and hence on the youngest zircons. Most of the concordant data-points lie between 230 and 250 Ma. Their calculated concordia age is 243.15 ± 1.3 Ma, i.e. Mid-Triassic, close to the boundary between the Anisian and the Ladinian. Figure 5c, d shows a comparison of obtained ages between the core (c) and the rim (d) for the population of the youngest zircons. A broad overlap between the two is apparent and they cannot be clearly distinguished by age, although most of the obtained rim ages appear slightly younger. A few older concordant

ages were obtained, with the oldest measured zircon showing approximate age around 3.1 Ga. The second oldest age originates from a 1.8 Ga old zircon core which is overgrown by 1.4 Ga old zircon. Three concordant ages of about 750 Ma were measured. Two of them originate from a single zircon crystal and one from a zircon core, which is overgrown by a Triassic zircon. CL images (Online Resource 6) generally indicate extensive compositional variations, which might not be fully resolved by the $30\ \mu\text{m}$ spots, limiting the accuracy of the calculated zircon age. The linear offset originating at a Mid-Triassic concordant age represents contamination with common lead. These data points were excluded from the calculation of the concordant age.

Massive sulphides and adjacent sediments

At Bobija, the ore constituents are fine-grained with pyrite being the most abundant sulphide (16B06, 16B07, 16B09_2) and generally present to a more than 60 vol%. While sphalerite, galena and tetrahedrite mainly vary in their respective

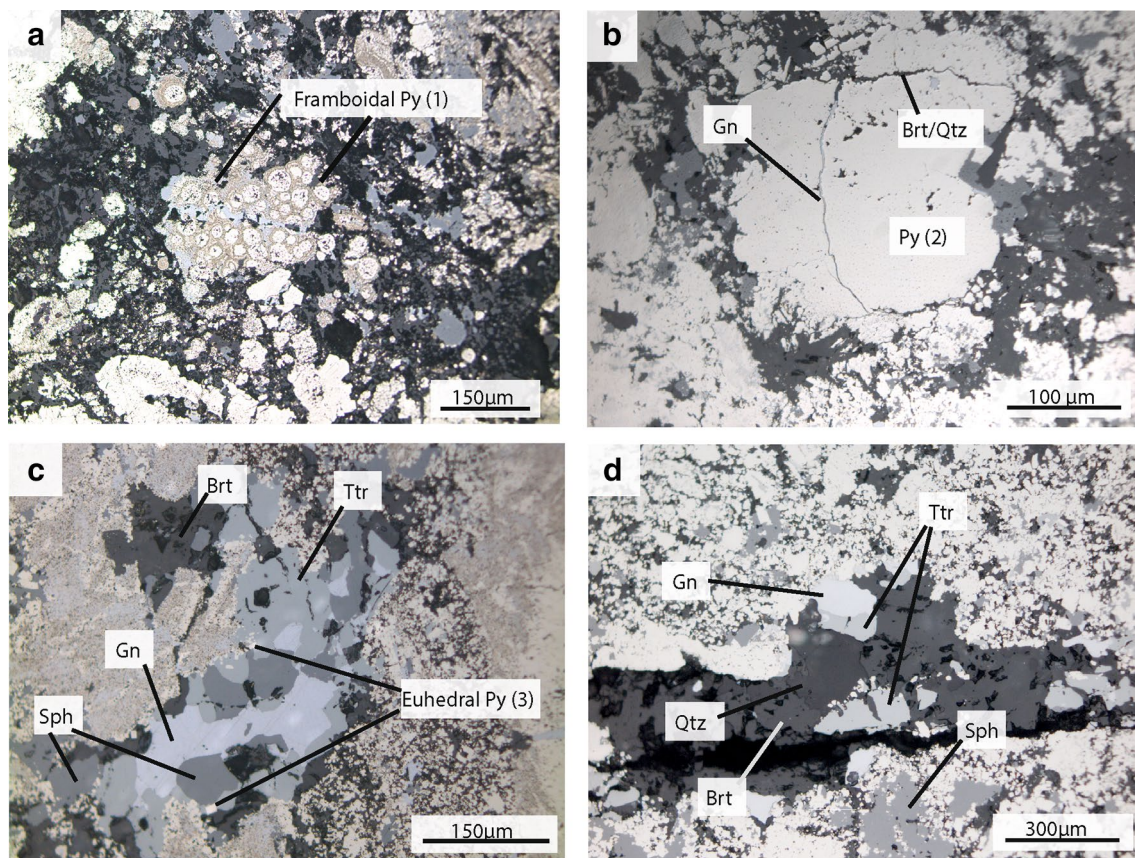


Fig. 6 Textures of sulphide minerals. **a** Framboidal pyrite with altered rims and galena between individual grains (Py1). **b** Crack within coarser, subhedral pyrite (Py2) filled with galena (Gn) and

barite (Brt)-quartz (Qtz). **c** Close intergrowth of galena, sphalerite (Sph), tetrahedrite (Ttr) and small, euhedral pyrite (Py3). **d** Close-up of a barite-quartz vein

grain sizes throughout the samples, several pyrite textural types can be distinguished: framboidal Py1 (20–50 µm), fractured and rather large subhedral/anhydral-Py2 (100–600 µm) and smaller euhedral pyrite-Py3 (<5 µm).

Framboidal Py1 type generally occurs as variably altered aggregates. In-between grains, galena is oftentimes present. Occasionally framboidal pyrite is associated with colloform textured sulphides. Py2 type grains are generally fractured and cracks are mostly filled with barite, quartz or both, but also appear filled with galena (Fig. 6b). Other sulphides are present in-between the pyrite grains, but not strongly intergrown with Py2 type. Py3 type is closely associated with subhedral sphalerite, galena and tetrahedrite. In this association, they dominate in their proportions over pyrite (Fig. 6c), displaying a strong mutual intergrowth (e.g. galena in sphalerite). Py2 and Py3 types cannot always be clearly distinguished in the samples. They are used as end-members to illustrate the presence of coarser grained, pyrite-dominated areas and finer-grained sphalerite, galena and tetrahedrite dominated areas in the samples, with usual gradation between the two end-members.

Barite and quartz are also irregularly distributed throughout the massive sulphides (<5 vol%), either as sole quartz veins or veins with a mixture of barite and quartz, both crosscutting the massive sulphide rock. Adjacent to the massive sulphides lie hydrothermally altered and dominantly silicified sediments (samples 16B14; 16B15). The most proximal ones (sample 16B15) contain significant amounts of pyrite (60–70 vol%) that is subhedral and coarse-grained, similar to the type Py2, sporadically resembling the framboidal Py1 type. Notable are the low amounts of other sulphides being present, where sphalerite and galena are << 1 vol% and tetrahedrite is absent.

We performed trace element analyses of sphalerite in the massive sulphide samples (16B06, 16B07, 16B09_2). Median trace element concentrations in sphalerite are shown in Table 1. Trace element data for single spots are given in Online Resource 7. Iron is a common element in natural sphalerite and can range from trace levels to up to 15 wt% (Cook et al. 2009). Obtained Fe contents in Bobija sphalerites are in the range from 1000 to 1600 µg/g. Mn concentrations in sphalerite are very low with values of

Table 1 Trace element concentrations in sphalerite

Sample	Mn	Fe	Co	Ni	Cu	Ga	Ge	As	Se	Ag
16B06 (9)	13.9	1170	bdl	0.18	238	6.05	0.41	12.9	14.7	16.8
SD	9.17	94.2	–	0.13	80.8	1.88	0.21	5.69	3.57	4.44
16B07 (17)	7.82	1200	0.11	0.13	587	14.9	1.44	37.2	22.4	34.6
SD	4.16	143	0.02	0.07	542	53.7	5.75	12.9	14.8	19.6
16B09_2 (11)	14.1	1450	bdl	0.57	534	8.64	2.30	42.0	12.9	23.9
SD	4.17	118	–	–	513	18.6	2.61	17.1	3.67	13.3
Sample	Cd	In	Sn	Sb	Au**	Hg*	Tl	Pb	Bi*	
16B06 (9)	838	0.58	1.67	9.21	0.11	2538	0.25	69.2	0.02	
SD	17.0	0.15	19.9	6.05	0.10	268	0.22	127	0.01	
16B07 (17)	928	1.11	255	44.9	0.10	2744	0.38	79.5	0.03	
SD	53.4	1.49	427	26.9	0.05	253	0.14	31.1	0.01	
16B09_2 (11)	972	2.08	227	80.7	0.07	4465	0.96	91.0	0.03	
SD	35.1	2.72	795	106	0.03	564	0.41	135	0.02	

The number in brackets behind the sample name represents the number of analyses spots from which the median was taken. Mul-ZnS-1 was used as a calibration material for most elements. For elements marked with * USGS MASS-1 and with ** NIST SRM 610 were used as a calibration material. See text for further details. Values and respective standard deviations (SD) are given in $\mu\text{g/g}$

5–39 $\mu\text{g/g}$ (median $\sim 11 \mu\text{g/g}$). All samples comprise rather low Cu contents (median: $\sim 400 \mu\text{g/g}$). Minor elevations possibly originate from tetrahedrite inclusions. Enrichment in Cu coincides with elevated Sn contents and could indicate the presence of kesterite ($\text{Cu}_2(\text{Zn,Fe})\text{SnS}_4$) nanoscale-exsolutions in sphalerite. Cadmium contents in sphalerite usually range from 0.1 to 0.5 wt% (Cook et al. 2009), whereas Bobija sphalerites fall in the lower end of this range (810–1010 $\mu\text{g/g}$). Silver contents are usually below 100 $\mu\text{g/g}$ in natural sphalerites, despite occurring as a solid solution, as well as micro-inclusions (Kelley et al. 2004b; Cook et al. 2009). 16B07 yields the highest Ag content (median 35 $\mu\text{g/g}$), 16B09_2 contains less (median: 24 $\mu\text{g/g}$) and 16B06 the lowest (median: 17 $\mu\text{g/g}$). Sphalerite is also seen as one of the main Hg-carrying sulphide minerals (Cook et al. 2009). Sphalerites from Bobija reach up to 5210 $\mu\text{g/g}$ Hg. The sample 16B09_2 is richest in Hg (median: 4465 $\mu\text{g/g}$), while 16B07 and 16B06 are significantly lower (median: 2744 and 2538 $\mu\text{g/g}$). Time-resolved signals during LA-ICPMS analysis further indicate inclusions of Hg bearing minerals, most probably cinnabar. Gold is uncommon in sphalerites and not incorporated in the crystal structure. This is reflected by low Au contents throughout the Bobija sphalerites (median: 0.1 $\mu\text{g/g}$). Gallium contents of the samples show a broad range from 2 to 176 $\mu\text{g/g}$. Its enrichment appears to correlate with elevations in Ge (0.4–19 $\mu\text{g/g}$). Indium contents are usually below 4 $\mu\text{g/g}$. Lead, Sb and Bi are regarded to be not incorporated in the crystal lattice, but to be present as micro-inclusions (Cook et al. 2009). Given the strong intergrowth of sulphides at Bobija, possible inclusions originate from other associated sulphides. Variations

in Pb concentrations might, therefore, originate from galena inclusions and intergrowth, whereas antimony that correlates with As and Cu indicates the presence of tetrahedrite.

The obtained element concentrations are used to calculate the sphalerite formation temperatures from the massive sulphide samples (GGIMFis geothermometer; Frenzel et al. 2016). The calculated temperatures are $152 \pm 18^\circ\text{C}$ (16B07), $179 \pm 14^\circ\text{C}$ (16B06) and $167 \pm 15^\circ\text{C}$ (16B09_2) (Table 3). The errors represent analytical errors only. Note that Frenzel et al. 2016 give a calibration error of about $\pm 50^\circ\text{C}$ for the GGIMFis thermometer.

We performed the sulphur isotope analysis on a mixed pyrite sample originating from Py2-type grains of the three massive sulphide samples (16B06, 16B07, 16B09_2). Additionally, we analysed barite from the samples 16B06, 16B07 and from a pure barite vein (16B12). The results of the sulphur isotope analysis are displayed in Table 2. $\delta^{34}\text{S}$ ratios of barites vary over a narrow range from +21.8 to +23.2‰. In contrast to the barites, the $\delta^{34}\text{S}$ of the mixed pyrite sample yields a negative value of -6.26‰ .

Table 2 Results of sulphur isotope analysis in barite and pyrite

Sample	$\delta^{34}\text{S}$	Description
16Py	−6.26	Mixture of pyrites from 16B06, 16B07 and 16B09_2 gained by micro drilling
16B06Ba	+22.3	Barite gained by micro drilling
16B07Ba	+21.8	Barite gained by micro drilling
16B12Ba	+23.2	Barite from barite vein within massive sulphides

Values are given in ‰ relative to V-CDT

Discussion

In the following, the geochemical and geochronological results for the volcanic rocks are combined with the information obtained by analyses of the sulphides. Finally, the genesis of the Bobija deposit and its geotectonic evolution is discussed.

Constraints on the tectonic setting derived from the rocks of the volcano-sedimentary succession

The ore mineralization at Bobija is sandwiched between Triassic siltstones and Permian black shales in the lower strata, and the volcanic rocks in the upper strata (Fig. 3). The youngest age obtained from zircons within the volcanoclastic sandstone sample 16B10 is a Mid-Triassic concordant age of 243.2 ± 1.3 Ma (Anisian). This delimits the volcanic activity to a Triassic advanced-rifting stage (Palinkaš et al. 2008a) in the Balkan Peninsula, preceding the opening of Neotethys in the late Anisian (Gawlick et al. 2012). This is confirmed by the presence of Neoproterozoic and Mid-Archean zircons, which testify for volcanic activity within a continental setting. Note that the obtained age does not necessarily represent the true age of the ore mineralization itself. Given the strong alteration of sediments directly overlying the mineralization, we expect the mineralization to be of a slightly younger age and hence to have been emplaced within an already existing host rock and to have altered it.

The continental rift setting is supported by the geochemistry of the lava sample. Applying the discrimination diagram after Meschede (1986) (Fig. 4b) the basaltic sample 16B13 is identified as within-plate alkaline basalt. Incompatible trace element patterns (Fig. 4c, d) are distinct from the mid-ocean ridge and subduction-related basalts. Th and U are > 10 times more enriched than in the MORB. La/Sm ratios of the Bobija volcanic rocks are > 4.5, whereas for MORB it is close to or just above 1. The absence of Nb and Ta negative anomalies further excludes them to originate from a subduction-zone setting (e.g. Allègre et al. 1977; Briquieu et al. 1984).

Compared with Triassic rift and Jurassic back-arc basalts from Vareš (Central Bosnia) and the Bükk area (Hungary) (Kiss et al. 2016a; Fig. 4), the trace element patterns of the Bobija volcanic and volcanoclastic sediments clearly lie within the array of Triassic basalts and are distinct from Jurassic ophiolitic basalts, again confirming a Triassic rift setting. The Bükk and Vareš regions are further known for their SEDEX type ore occurrences (Palinkaš et al. 2008a, 2016; Kiss et al. 2016b). These similarities could imply a genetical link between the geological units and metallogenic districts. This will be discussed further below.

Constraints derived from the sulphide mineralization

Textural characteristics and a sequence of mineralization stages

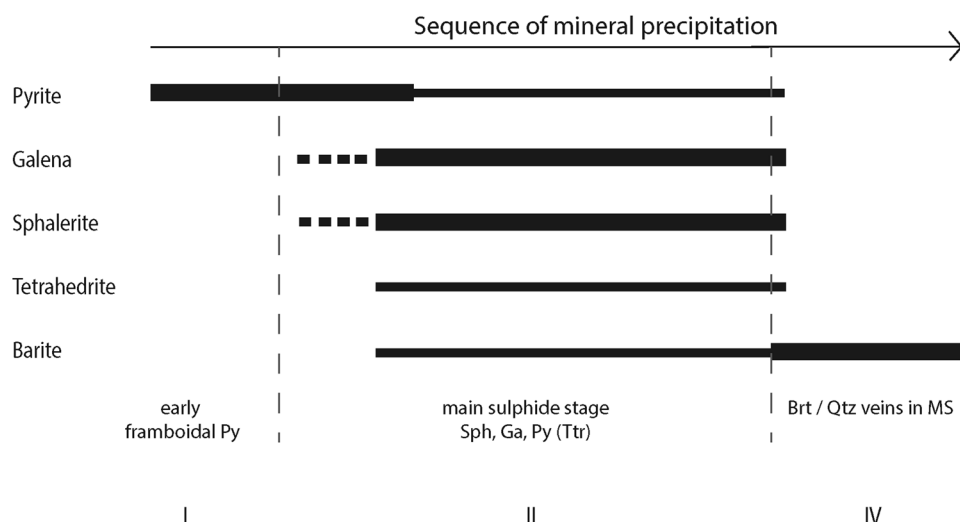
The formation of framboidal pyrite in basin sediments can be syngenetic (formed within the water column) and diagenetic (formed within the sediments) (e.g. Raiswell and Berner 1985; Wilkin et al. 1996). It can also be found in the paragenesis of metalliferous ore deposits (e.g. Halbach et al. 1993; Wilkin and Barnes 1997). The framboidal Py1 type at Bobija is only found as scarce relicts. It is expected to have been overprinted by later hydrothermal sulphide deposition and further obliterated by continuous diagenesis and brecciation. It is unclear whether they are remnants of former sedimentary host rock, or they represent the earliest expression of an arising hydrothermal ore system. Wilkin and Barnes (1997) concluded that framboidal pyrite offers nucleation surfaces for secondary pyrite growth, eventually leading to euhedral forms of pyrite. Similarly, the signs of secondary overgrowth are also observed in the Bobija massive sulphide samples, as witnessed by the partially observed altered rims of the framboidal pyrites and the presence of galena in-between single framboidal pyrite grains (Fig. 6a).

The coarser grained subhedral Py2 type and the fine-grained euhedral Py3 type are obviously generated within a hydrothermal system. Where it appears, the Py2 type is modally dominant, whereas, in the samples where Py3 type occurs, it is subordinate, associated and intergrown with sphalerite, galena and tetrahedrite, as previously reported by Radosavljević et al. (2013b). The dominant part of the sulphides displays textures with gradations between the two end-members. The fractured appearance of Py2 type and the lack of fracturing in other sulphides implies that this pyrite type is relatively older than Py3 type and coeval with other sulphides. This is confirmed by filling of other ore minerals (galena and/or barite) within the cracks of Py2 type (Fig. 6b). A similar observation may be acquired from sample 16B15, which borders the massive sulphides and is nearly devoid of sphalerite, galena and tetrahedrite (< 1 vol%).

In summary, we interpret the observed ore textures as being generated during different stages throughout the same hydrothermal system.

The timing of barite deposition is two-fold. It is not only restrained to layers or veins but is also subordinately deposited within the massive sulphide mineralization, arguing towards a partially coeval deposition with the sulphides of the main sulphide stage. As for the barite and barite/quartz veins, they seem to be the expression of a later stage, post-dating the massive sulphide mineralization. Here, quartz and barite veins cut the massive sulphides. In sample 16B06 a

Fig. 7 Sequence of mineralization of ore minerals based on the reflected light investigation. The thickness of the bars represents the relative abundance of the respective mineral



quartz-barite vein is cut by a barite vein and hence the former is relatively older. Nevertheless, this was only observed in one place and needs further investigation. Large pure barite veins are also irregularly distributed throughout the massive sulphides. A possible explanation for the vein hosted barite could be infiltration of seawater along rift-induced faults.

Based on the above discussion, a sequence of relative mineralization stages is proposed and shown in Fig. 7. Note that this sequence is only preliminary due to a limited number of investigated samples. Three main stages are distinguished: Stage I: Early framboidal pyrite mineralization (Py1 type); Stage II: Hydrothermal sulphide deposition; main sulphide stage (Py2/Py3 type + varying modal proportions of galena, sphalerite, tetrahedrite and barite); Stage III: Barite/barite-quartz veining.

The timing and the mode of emplacement of the mineralization are expected to be syn- to post-diagenetic. As mentioned above the framboidal pyrite (Py1) could be syngenetic, diagenetic or be the earliest expression of a hydrothermal system. Relative to framboidal Py1, the main sulphide stage is clearly younger and displays emplacement with changing modal proportions of mineralized sulphides over time. The underlying feeder zone implies that the sulphides are for the most part younger than the host rocks with continuous sulphide deposition to the seafloor.

Given the temporal and geochemical similarities of volcanic rocks to ones from SEDEX deposits in Bosnia (“Constraints on the tectonic setting derived from the rocks of the volcano-sedimentary succession” section) the question arises whether the observed textures are consistent with a SEDEX type deposit, which supposedly entails an exhalative, synsedimentary sulphide deposition. While the majority of SEDEX deposits are bedded ores distal to the vent-complex, deposits proximal or within the vent-complex are characterized by massive to weakly bedded massive sulphide

ores (Lydon 1996; Emsbo et al. 2016). Above that, while the bedding in SEDEX deposits used to be interpreted as the expression of exhalation to the seafloor, recent studies suggest that bedded textures in SEDEX deposits classically interpreted to be of exhalative origin, can form by replacement of non-consolidated sediments during and after diagenesis, without the necessity for an exhalative deposition (Kelley et al. 2004a, b; Reynolds et al. 2015; Magnall et al. 2015). We discuss the deposit type in more detail below.

Sphalerite formation temperatures

Major element composition was traditionally utilized for discrimination of sphalerites typical for different deposits (Qian 1987). In a more recent study, Frenzel et al. (2016) developed a new geothermometer based on a statistical analysis of published sphalerite data. They conducted a principal component analysis which demonstrated that the mean concentrations of Fe, Ga, Ge, In and Mn vary according to deposit type. They found that these differences are one-dimensional and that they can be expressed by a single-number (PC1*). This number strongly correlates with homogenization temperature of fluid inclusions, leading them to propose a new sphalerite geothermometer (GGIMFis).

The calculated temperatures of 152 ± 18 °C (16B07), 179 ± 14 °C (16B06) and 167 ± 15 °C (16B09_2) (Table 3) show minor variance and are consistent with each other (note that the errors are only analytical errors, whereas

Table 3 Calculated median temperatures after the GGIMFis sphalerite geothermometer (Frenzel et al. 2016)

	16B07	16B06	16B09_2
PC1*	1.04	0.53	0.75
T (°C)	152 ± 18	179 ± 14	167 ± 15

the calibration error of the GGIMFis thermometer is about ± 50 °C). They are in good agreement with temperatures depicted for the Bosnian SEDEX deposits in the Vareš region (110–230 °C; Palinkaš et al. 2003, 2008a) and SEDEX deposits worldwide (100–200 °C; Gardner and Hutcheon 1985; Leach et al. 2004, 2005; Lydon et al. 2004; Polito et al. 2006).

Constraints derived from the sulphur isotopes

As presented earlier (Table 2), the $\delta^{34}\text{S}$ values of the three analysed barite samples vary from +21.8 to +23.2‰ and the mixed pyrite sample gave a $\delta^{34}\text{S}$ value of −6.3‰. Obtained $\delta^{34}\text{S}$ values are displayed in Fig. 8 together with typical $\delta^{34}\text{S}$ values for evaporites, magmatic rocks, sedimentary pyrite and seawater.

$\delta^{34}\text{S}$ values of barite are in good agreement with data presented by Šiftar (1988), who gave a mean value of +23‰ for Triassic seawater and is in accordance with recently published high-resolution sulphur isotope record based on Late Permian and Triassic evaporites (Bernasconi et al. 2017). This value is in sharp contrast with the values of $\sim +10$ ‰ of barite from Permian deposits of the Dinarides (e.g. Ljubija, Palinkaš et al. 2016) formed during early intracontinental Tethyan rifting, but is comparable with $\delta^{34}\text{S}$ values for barite from the Triassic Vareš deposit in Central Bosnia (+20.7 to +28.6‰) and Veovača (+20.89‰) (Šiftar 1988).

On the other hand, $\delta^{34}\text{S}$ of pyrite at Vareš yield values of −11‰ (Šiftar 1988), which is comparable with $\delta^{34}\text{S}$ in pyrite from Bobija (−6.3‰). The negative $\delta^{34}\text{S}$ value in pyrite indicates a reduced sulphur source, either originated from bacteriogenically/thermochemically reduced seawater sulphate or derived from the ore-forming fluids themselves. For the Vareš and Veovača SEDEX deposits in Central Bosnia, Palinkaš et al. (2003) demonstrated that seawater had an important role, both as a possible fluid source and as a sulphur source.

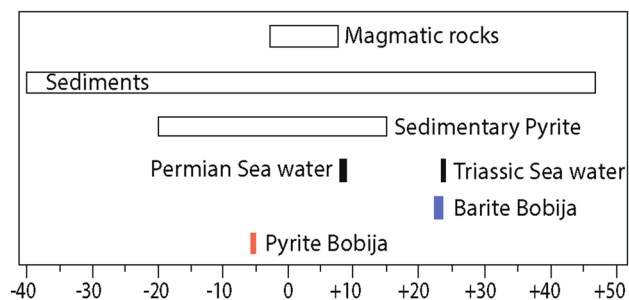


Fig. 8 Sulphur isotopes ($\delta^{34}\text{S}$ in ‰ relative to V-CDT) of barite and pyrite are given in comparison with typical $\delta^{34}\text{S}$ values for magmatic rocks (Hoefs 1997), sedimentary pyrite (Farquhar et al. 2010), Permian and Triassic sea water (Šiftar 1988) are shown. Sedimentary pyrite hereby includes framboidal pyrite (Schoonen 2004)

Generally, the extent of the isotope fractionation depends on the ultimate sulfur source (Triassic sea water), available organic contents, amount of sulphate reducing bacterias, salinity, temperature, etc. (Kelley et al. 2004b). When sulphate reduction is achieved by bacterial activity, the ranges in fractionation larger than 15–25‰ are obtained, whereas in the case of the thermal sulphate reduction, isotope fractionations of 15–25‰ are expected (Ohmoto and Goldhaber 1997). At Bobija the isotopic fractionation is > 29 ‰ between Triassic seawater (~ 23 ‰) and pyrite (−6.3‰) indicating that at least some of the sulphur originates from the bacterial reduction of seawater within organic-rich host sediments.

Classification of the Bobija deposit: implications for metal and fluid source and regional geology

According to the above findings, we conclude that the Bobija deposit is sediment-hosted and associated with volcanic rocks from the advanced Tethyan rifting phase. It is underlain by a feeder zone and the mineralization displays a massive sulphide texture. The sulphides were dominantly deposited syndiagenetically within or proximal to the vent-complex to the sub-seafloor. The temperature of the metal-bearing fluid was approximately 150–180 °C. The dominant source of sulphur was most probably reduced seawater sulphate.

These characteristics are in general agreement to recent SEDEX deposit models (e.g. Emsbo et al. 2016 and references therein). As pointed out by Leach et al. (2005) the term ‘SEDEX’ is based upon a genetical criterion that implies an exhalative, synsedimentary deposition of the ore, although some SEDEX deposits only mimic exhalative textures, while they were formed by replacement processes below the sea floor (Kelley et al. 2004a, b; Reynolds et al. 2015; Magnall et al. 2015, 2016). Leach et al. (2005), therefore, proposed more descriptive term ‘clastic-dominated’ to be generally used instead of SEDEX, as having no genetic connotation. Clastic-dominated Pb–Zn ores are hosted in shale, sandstone, siltstone, or mixed clastic rocks. They occur in various tectonic settings such as passive margins, back-arcs and continental rifts, and sag basins. According to this classification, we describe the Bobija deposit as ‘clastic-dominated’ Pb–Zn deposit which formed within a rift-tectonic setting.

Following this classification, the metal source can be inferred to be continental clastic sediments within a rift basin, as it has been demonstrated by radiogenic and stable isotopic compositions for other sediment-hosted Pb–Zn ores (Ayuso et al. 2004; Emsbo and Johnson 2004; Large et al. 2005; Leach et al. 2005). Fluid solute data and relation between mineralization and evaporation of seawater on basin platforms indicate that residual brines from

seawater evaporation on basin platforms are the main fluid source in sediment-hosted Pb–Zn deposits (Leach et al. 2005; Emsbo 2009 and references therein). This could imply a link between the Mid-Triassic carbonate formations of the External Dinarides and Pb–Zn deposits of the advanced-Tethyan rifting stage. Given the similarities to the Bobija deposit, a similar fluid source seems likely. Even so, additional fluid inclusion investigation is necessary to unequivocally pinpoint the origin of ore-forming fluids at Bobija.

Given the presence of the feeder zone underlying the deposit and the close association with volcanic rocks, it is debatable whether the Bobija deposit could also be described as a Volcanic-Hosted Massive Sulphide (VHMS) type deposit. VHMS deposits form by processes arguably comparable to sediment-hosted Pb–Zn deposits. Critical is the exact interplay between volcanic activity and mineralization. In VHMS, the volcanic activity is the driving force of the hydrothermal system, the major sulphur source is formed by reduced seawater sulphate, while the metals are derived by seawater convection and leaching underlying volcanic pile and basement rocks (Large 1992, Huston et al. 2010). Therefore, future studies at Bobija should improve the resolution of the absolute age dating by involving the Re–Os dating of pyrite that would give important insight into the temporal relationship between the volcanic activity and mineralization. Nevertheless, the proposed continental rift regime is generally favourable for sediment-hosted Pb–Zn deposits rather than for VHMS (Huston et al. 2010).

The classification as clastic-dominated Pb–Zn deposit within a rift-tectonic setting is also in accordance with the occurrence of SEDEX deposits in Central Bosnia (Palinkaš et al. 2008a, 2016) and the Darnó Unit in the Bükk Unit (Kiss et al. 2016b). This supports the close relationship between tectonic setting and deposit type in the Balkan Peninsula. The above findings underline the stratigraphical similarities between the Bükk Unit and the Jadar Block as previously described by Filipović et al. (2003). The Jadar Block hosting the Bobija deposit, the East-Bosnian Durnator nappe hosting Central Bosnian SEDEX deposits and the Bükk Unit are all part of the more external zones of the Dinarides and are tectonically derived from Adria. While the exact geodynamic history of the Jadar Block is still a matter of ongoing debate (e.g. Palinkaš et al. 2008a; Schmid et al. 2008; Robertson et al. 2009), the observed similarities support a common origin of the Jadar Block and more external zones of the Dinarides which contradicts a separate evolution (Schmid et al. 2008).

Model of metallogenesis

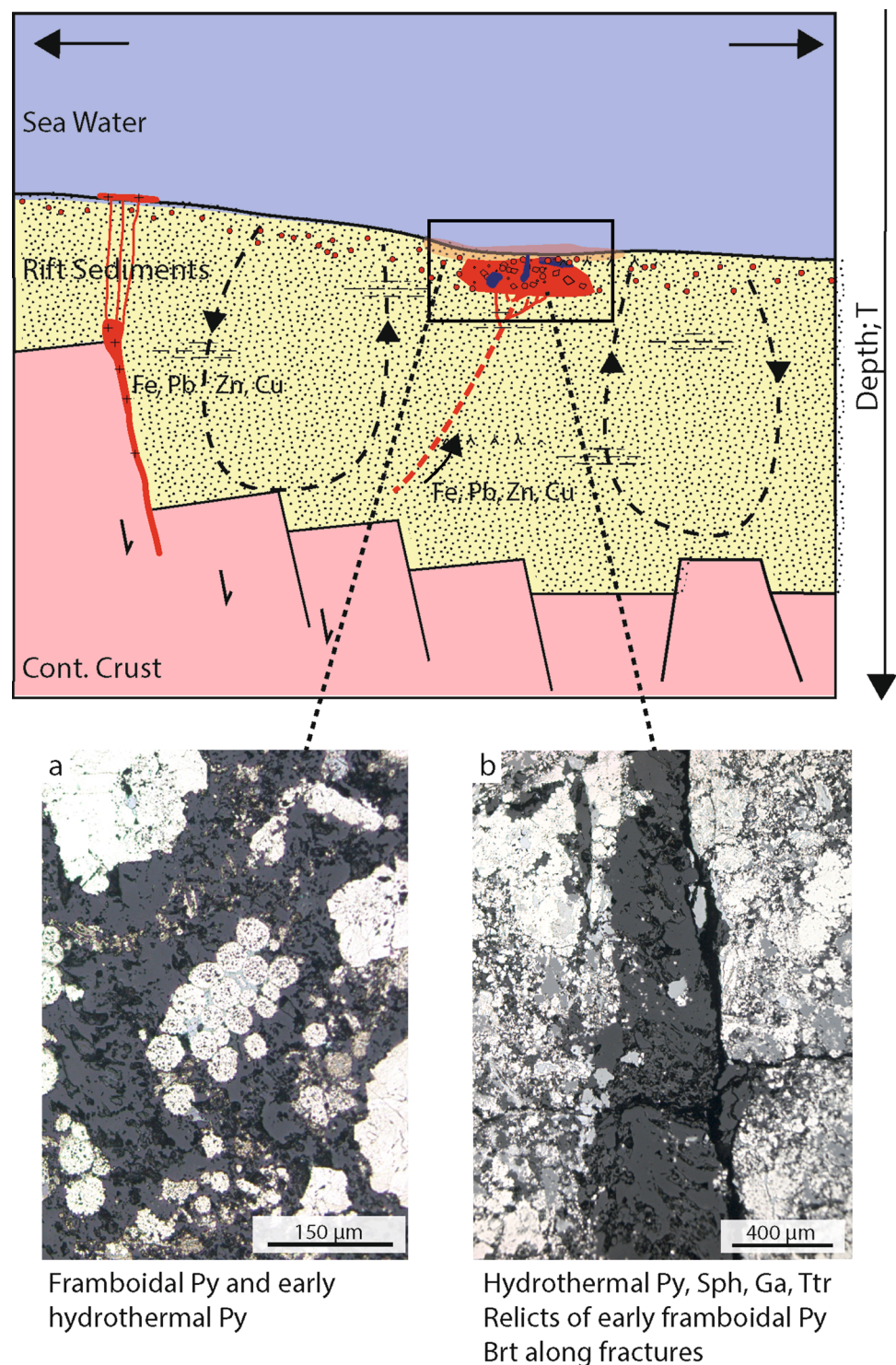
Based on the characteristics discussed above, a model sketch for the genesis of the Bobija deposit is proposed in Fig. 9. It illustrates the inferred main ore-forming processes related to the emergence of a hydrothermal system within a Triassic rift basin at the distal north-eastern edge of Adria. We assume that the metals are mobilized from the underlying sediment formations (Triassic rift sediments, Permian black shales) and transported upwards along faults to the sub-sea-floor. The metal bearing brine precipitated pyrite, together with sphalerite, galena and tetrahedrite (Fig. 9b). Ore textures might further indicate an early pyrite-dominated hydrothermal phase. Barite is interpreted to have formed coevally with the main sulphide stage and to have incorporated seawater sulphate. During continuous rifting, it enters the massive sulphide mineralization along faults and precipitates within the massive sulphide mineralization.

Conclusion

The Bobija deposit is a polymetallic sulphide deposit, consisting mainly of pyrite, lesser amounts of sphalerite, galena, tetrahedrite and associated barite. Textures in sulphides are the result of different stages throughout a prevailing hydrothermal system. In the footwall, a feeder zone is present, situated within Triassic sediments. Maximum Mid-Triassic age is indicated by U–Pb analysis of zircons from a volcanoclastic sandstone sample in the uppermost strata.

The Bobija deposit resembles characteristics generally referred to as SEDEX-type, in agreement with similar deposits occurring in the Dinarides. Given the reservations made by Leach et al. (2005), their proposed more descriptive terminology is followed, hereby classifying the deposit as a sediment-hosted, clastic-dominated Pb–Zn deposit within an advanced intra-continental rift setting. Characteristics are temperatures of 150–180 °C and a setting within a Triassic (243.1 Ma) rift-regime. A syn-rift hydrothermal system deposited pyrite, sphalerite, galena and tetrahedrite. The strong intergrowth of sulphides argues towards a mainly coeval deposition during the main sulphide stage. The presence of tetrahedrite, massive ore textures and an underlying feeder zone, depict a dominantly vent-proximal location of the massive sulphide mineralization. Triassic rifting is believed to have initiated faulting and possibly delivered heat, thereby facilitating fluid flow in the rift basin. This led to metal extraction and subsequent deposition on and beneath the seafloor, proximal to or within the hydrothermal vent-complex. Above findings are in good agreement with similar deposits

Fig. 9 Model sketch for the genesis of the Bobija deposit, together with matching thin section photographs. The Triassic rift regime leads to the formation of a rift basin, which at an early stage is filled with continental rift and increasingly with marine sediments. The intracontinental rifting environment is supported by the heterogeneous age populations in the volcanoclastic sandstone sample (16B10). The rifting delivered an elevated geotherm and induced faulting, both facilitating fluid circulation within the basin and initiating the emergence of the hydrothermal system. Fluids leach Fe, Pb, Zn and Cu from the underlying sediments. The dissolved metals are deposited to the sub- and possibly to the sea floor as part of the vent-complex. At an early stage dominantly pyrite is deposited (a), while at a later main sulphide stage pyrite, sphalerite, galena, tetrahedrite and barite are deposited (b). Preserved is mainly the massive part proximal to and/or within the vent. Barite might have formed a cap above the massive sulphides and infiltrated the massive sulphide mineralization along faults (b)



from Central Bosnia, which further underlines common origin of the Jadar Block and more external zones of the Dinarides.

Acknowledgements The company Balkan Exploration and Mining is cordially thanked for allowing us to sample the outcrops and for logistics and expert support. This work was supported through the

Serbian Ministry of Education, Science and Technological Development, Grant/Award Number: 176016. We would like to thank the speleothem research group of the University of Mainz for their support with the micro mill. Dr. Moreno from the University of Barcelona is acknowledged for the analyses of the sulphur isotopes. Kristina Šarić and Vladica Cvetković are thanked for useful discussions and critical comments on an earlier version of the manuscript.

References

- Allègre CJ, Treuil M, Minster J-F, Minster B, Albarède F (1977) Systematic use of trace element in igneous process. *Contrib Miner Petrol* 60:57–75. <https://doi.org/10.1007/BF00372851>
- Ansorge J, Blundell D, Mueller S (1993) Europe's lithosphere-seismic structure. In: Blundell D, Freeman R, Mueller S (eds) *A continent revealed: the European geotraverse*. Cambridge University Press, pp 33–69
- Ayuso RA, Kelley KD, Leach DL, Young LE, Slack JF et al (2004) Origin of the red dog Zn–Pb–Ag deposits, Brooks Range, Alaska: evidence from regional Pb and Sr isotope sources. *Econ Geol* 99:1533–1553
- Bernasconi SM, Meier I, Wohlwend S, Brack P, Hochuli PA, Bläsi H, Wortmann UG, Ramseyer K (2017) An evaporite-based high-resolution sulfur isotope record of Late Permian and Triassic seawater sulfate. *Geochim Cosmochim Acta* 204:331–349
- Berza T, Constantinescu E, Vlad S-N (1998) Upper Cretaceous magmatic series and associated mineralisation in the Carpathian–Balkan Orogen. *Resour Geol* 48:291–306
- Briqueu L, Bougault H, Joron JL (1984) Quantification of Nb, Ta, Ti and V anomalies in magmas associated with subduction zones: petrogenetic implications. *Earth Planet Sci Lett* 68:297–308. [https://doi.org/10.1016/0012-821X\(84\)90161-4](https://doi.org/10.1016/0012-821X(84)90161-4)
- Cook NJ, Ciobanu CL, Pring A, Skinner W, Shimizu M et al (2009) Trace and minor elements in sphalerite: a LA-ICPMS study. *Geochim Cosmochim Acta* 73:4761–4791
- Cvetković V et al (2016) Geology of South-Eastern Europe. In: Papić O (ed) *Mineral and thermal waters of Southeastern Europe*. Springer, Berlin, pp 1–29
- Dimitrijević MD (1992) Geological atlas of Serbia, 1: 2.000.000. Republic foundation of geological investigations and Geological Survey Gemini
- Emsbo P (2009) Geologic criteria for the assessment of sedimentary exhalative (sedex) Zn–Pb–Ag deposits. US Geological Survey open-file report, 1209, 21
- Emsbo P, Johnson CA (2004) Formation of modern and Paleozoic stratiform barite at cold methane seeps on continental margins—comment. *Geology* 32:e64
- Emsbo P, Seal RR, Breit GN, Diehl SF, Shah AK (2016) Sedimentary exhalative (sedex) zinc-lead-silver deposit model. U.S. Geological Survey scientific investigations report, 2010-5070-N
- Farquhar J, Wu N, Canfield DE, Oduro H (2010) Connections between sulfur cycle evolution, sulfur isotopes, sediments, and base metal sulfide deposits. *Econ Geol* 105:509–533
- Filipović I, Jovanović D, Sudar M, Pelikán P, Kovács S et al (2003) Comparison of the Variscan-Early Alpine evolution of the Jadar Block (NW Serbia) and “Bükkium” (NE Hungary) terranes; some paleogeographic implications. *Slovak Geol Mag* 9:3–21
- Frenzel M, Hirsch T, Gutzmer J (2016) Gallium, germanium, indium, and other trace and minor elements in sphalerite as a function of deposit type—a meta-analysis. *Ore Geol Rev* 76:52–78
- Gardner HD, Hutcheon I (1985) Geochemistry, mineralogy, and geology of the Jason Pb–Zn deposits, Macmillan Pass, Yukon, Canada. *Econ Geol* 80:1257–1276
- Gawlick H-J, Goričan Š, Missoni S, Lein R (2012) Late Anisian platform drowning and radiolarite deposition as a consequence of the opening of the Neotethys ocean (High Karst nappe, Montenegro). *Bulletin de la Société géologique de France* 183:349–358
- Halbach P, Pracejus B, Maerten A (1993) Geology and mineralogy of massive sulfide ores from the central Okinawa Trough, Japan. *Econ Geol* 88:2210–2225
- Haas J, Kovács S, Krystyn L, Lein R (1995) Significance of late Permian-Triassic facies zones in terrane reconstructions in the Alpine-North Pannonian domain. *Tectonophysics* 242:19–40
- Heinrich CA, Neubauer F (2002) Cu–Au–Pb–Zn–Ag metallogeny of the Alpine–Balkan–Carpathian–Dinaride geodynamic province. *Miner Depos* 37:533–540
- Hoefs J (1997) *Stable isotope geochemistry*, vol 201. Springer, Berlin
- Huston DL, Pehrsson S, Eglinton BM, Zaw K (2010) The geology and metallogeny of volcanic-hosted massive sulfide deposits: variations through geologic time and with tectonic setting. *Econ Geol* 105:571–591. <https://doi.org/10.2113/gsecongeo.105.3.571>
- Janković S (1987) Genetic types and major Triassic deposits of the Dinarides, Yugoslavia. *Mineral deposits on the TEMB between the Alps and Pamirs. IGCP project*, vol 169, pp 11–33
- Janković S (1990) The ore deposits of Serbia (Yugoslavia): Regional metallogenic settings, environments of deposition and types. Republic Social Fund for Geological Investigations and Faculty of Mining and Geology, Department for Economic Geology, Belgrade, vol 159, pp 1–760
- Janković S (1997) The Carpatho-Balkanides and adjacent area: a sector of the Tethyan Eurasian metallogenic belt. *Miner Depos* 32:426–433
- Jelenković R, Kostić A, Životić D, Ercegovac M (2008) Mineral resources of Serbia. *Geol Carpatia* 59(4):345–361
- Karamata S (2000) Mineralization related to the Triassic rifting in the Borovic-Vareš-Čevljanovići-Kalinovik zone (Bosnia). *Acta Geol Hung Bp* 43:15–23
- Karamata S, Knežević V, Cvetković V (2000) Petrology of the Triassic basaltoid rocks of Vareš (Central Bosnia). *Acta Geol Hung* 43:1–14
- Kelley KD, Dumoulin JA, Jennings S (2004a) The Anarraaq Zn–Pb–Ag and barite deposit, northern Alaska: evidence for replacement of carbonate by barite and sulfides. *Econ Geol* 99:1577–1591
- Kelley KD, Leach DL, Johnson CA, Clark JL, Fayek M et al (2004b) Textural, compositional, and sulfur isotope variations of sulfide minerals in the Red Dog Zn–Pb–Ag deposits, Brooks Range, Alaska: implications for ore formation. *Econ Geol* 99:1509–1532
- Kiss G, Molnár F, Palinkaš AL, Kovács S, Horvatić H (2012) Correlation of Triassic advanced rifting-related Neotethyan submarine basaltic volcanism of the Darnó Unit (NE-Hungary) with some Dinaridic and Hellenidic occurrences on the basis of volcanological, fluid-rock interaction, and geochemical characteristics. *Int J Earth Sci* 101:1503–1521
- Kiss G, Molnár F, Palinkaš AL (2016a) Hydrothermal processes related to Triassic and Jurassic submarine basaltic complexes in north-eastern Hungary and in the Dinarides and Hellenides. *Geol Croat* 69:39–64
- Kiss G, Oláh E, Zaccarini F, Szakáll S (2016b) Neotethyan rifting-related ore occurrences: study of an accretionary mélange complex (Darnó Unit, NE Hungary). *Geol Carpath* 67:105–115
- Knežević V, Cvetković V (2000) Triassic rifting magmatism of the Dinarides. In: Karamata S, Janković S (eds) *Proceedings of the international symposium “Geology and Metallogeny of the Dinarides and the Vardar zone”*. Academy of Science and Arts of the Republic of Srpska, Collections and monographs, Banja Luka - Serbian Sarajevo, vol 1, pp 149–160
- Large RR (1992) Australian volcanic-hosted massive sulfide deposits; features, styles, and genetic models. *Econ Geol* 87:471–510. <https://doi.org/10.2113/gsecongeo.87.3.471>
- Large RR, Bull SW, McGoldrick PJ, Walters SG (2005) Stratiform and strata-bound Zn–Pb–Ag deposits in Proterozoic sedimentary basins, northern Australia. *Econ Geol* 100:931–963
- Leach DL, Marsh E, Emsbo P, Rombach C, Kelley K et al (2004) Nature of hydrothermal fluids at the shale-hosted red dog Zn–Pb–Ag deposits, Brooks Range, Alaska. *Econ Geol* 99:1449–1480
- Leach DL, Sangster DF, Kelley KD, Ross RL, Garven G et al (2005) Sediment-hosted Pb–Zn deposits: a global perspective. *Econ Geol* 100:561–608

- Ludwig K (2012) User's manual for Isoplot version 3.75–4.15: a geochronological toolkit for Microsoft Excel. Berkeley Geochronology Center Special Publication, Berkeley
- Lydon JW (1996) Sedimentary exhalative sulphides (SEDEX). *Geol Can Miner Depos Types* 8:130–152
- Lydon JW, Deb M, Goodfellow WD (2004) Geology of the Belt-Purcell basin and the Sullivan deposit. Sediment-hosted lead-zinc sulfide deposits: attributes and models of some major deposits in India, Australia, and Canada. Narosa Publishing House, New Delhi, pp 100–148
- Magnall JM, Gleeson SA, Paradis S (2015) The importance of siliceous radiolarian-bearing mudstones in the formation of sediment-hosted Zn-Pb $\pm$ Ba mineralization in the Selwyn Basin, Yukon, Canada. *Econ Geol* 110:2139–2146. <https://doi.org/10.2113/econgeo.110.8.2139>
- Magnall JM, Gleeson SA, Stern RA, Newton RJ, Poulton SW et al (2016) Open system sulphate reduction in a diagenetic environment— isotopic analysis of barite ($\delta^{34}\text{S}$ and $\delta^{18}\text{O}$) and pyrite ($\delta^{34}\text{S}$) from the Tom and Jason Late Devonian Zn–Pb–Ba deposits, Selwyn Basin, Canada. *Geochim Cosmochim Acta* 180:146–163. <https://doi.org/10.1016/j.gca.2016.02.015>
- McDonough W, Sun S (1995) The composition of the Earth. *Chem Geol* 120:223–253
- Memović E, Cvetković VD, Knežević V, Zakariadze G (2003) The Triassic metabasalts of Dudin Krš, near Kosovska Mitrovica, Serbia. *Geološki Anali Balkanskog Poluostrva* 65:85–91
- Meschede M (1986) A method of discriminating between different types of mid-ocean ridge basalts and continental tholeiites with the Nb Zr Y diagram. *Chem Geol* 56:207–218
- Ohmoto H, Goldhaber MB (1997) Sulfur and carbon isotopes. In: Barnes HL (ed) *Geochemistry of hydrothermal ore deposits*, 3rd edn. Wiley, New York, pp 517–612
- Palinkaš AL et al (2003) Vareš and Veovča, Triassic, Fe–Pb–Zn–Ba SEDEX deposits, related to the advanced Tethyan rifting, Central Bosnia. In: Eliopoulos DG (ed) *Mineral exploration and sustainable development. Proceedings of the seventh biennial SGA meeting*. Millpress, Rotterdam, pp 1221–1225
- Palinkaš AL, Bermanec V, Borojević Šoštarić S, Kolar-Jurkoviček T, Strmić Palinkaš S et al (2008a) Volcanic facies analysis of a subaqueous basalt lava-flow complex at Hruškovac, NW Croatia—evidence of advanced rifting in the Tethyan domain. *J Volcanol Geotherm Res* 178:644–656
- Palinkaš AL, Borojević Šoštarić S, Strmić Palinkaš S (2008b) Metallogeny of the northwestern and central Dinarides and southern Tisia. *Ore Geol Rev* 34:501–520
- Palinkaš AL, Borojević Šoštarić S, Strmić Palinkaš S, Prochaska W, Pécskay Z et al (2016) The Ljubija geothermal field: a herald of the Pangea break-up (NW Bosnia and Herzegovina). *Geol Croat* 69:3–30
- Pamić J (1984) Triassic magmatism of the Dinarides in Yugoslavia. *Tectonophysics* 109:273–307
- Pearce J, Parkinson I (1993) Trace element models for mantle melting: application to volcanic arc petrogenesis. *Geol Soc Lond Spec Publ* 76:373–403
- Polito PA, Kyser TK, Golding SD, Southgate PN (2006) Zinc deposits and related mineralization of the Burketown mineral field, including the world-class Century deposit, northern Australia: fluid inclusion and stable isotope evidence for basin fluid sources. *Econ Geol* 101:1251–1273
- Protić Lj, Filipović I, Pelikan P, Jovanović D, Kovacs S, Sudar M, Hips K, Less G, Cvijić R (2000) Correlation of the Carboniferous, Permian and Triassic sequences of the Jadar block, Sana-Una and Bukkium terranes. In: Karamata S, Janković S (eds) *Proceedings of the international symposium “Geology and Metallogeny of the Dinarides and the Vardar Zone”*. Academy of Sciences and Arts of the Republic of Srpska, Collections and monographs, Banja Luka – Serbian Sarajevo, vol 1, pp 61–69
- Qian Z (1987) Trace elements in galena and sphalerite and their geochemical significance in distinguishing the genetic types of Pb–Zn ore deposits. *Chin J Geochem* 6:177–190
- Radosavljević SA, Stojanović JN, Radosavljević-Mihajlović AS, Kašić VD (2013a) Polymetallic mineralization of the Boranja orefield, Podrinje Metallogenic District, Serbia: zonality, mineral associations and genetic features. *Periodico di Mineralogia* 82:61–87
- Radosavljević SA, Vuković N, Stojanović J, Pačevski A et al (2013b) Polymetallic barite-sulfide deposit Bobija, Triassic volcanogenic-sedimentary zone of Podrinje, western Serbia: mineralogy and chemism of sulfides. In: *Proceedings of 45th international October conference on mining and metallurgy*, pp 531–534
- Radosavljević SA, Stojanović J, Pačevski A, Radosavljević-Mihajlović A (2016) A review of Pb–Sb (As)–S, Cu (Ag)–Fe (Zn)–Sb (As)–S, Ag (Pb)–Bi (Sb)–S and Pb–Bi–S (Te) sulfosalt systems from the Boranja orefield, West Serbia. *Geološki Anali Balkanskog Poluostrva* 2016:1–12
- Raiswell R, Berner RA (1985) Pyrite formation in euxinic and semi-euxinic sediments. *Am J Sci* 285:710–724. <https://doi.org/10.2475/ajs.285.8.710>
- Reynolds MA, Gingras MK, Gleeson SA, Stemler JU (2015) More than a trace of oxygen: ichnological constraints on the formation of the giant Zn–Pb–Ag $\pm$ Ba deposits, Red Dog district, Alaska. *Geology* 43:867–870. <https://doi.org/10.1130/G36954.1>
- Robertson A, Karamata S, Šarić K (2009) Overview of ophiolites and related units in the Late Palaeozoic–Early Cenozoic magmatic and tectonic development of Tethys in the northern part of the Balkan region. *Lithos* 108:1–36
- Schmid SM, Bernoulli D, Fügenschuh B, Matenco L, Schefer S et al (2008) The Alpine-Carpathian-Dinaridic orogenic system: correlation and evolution of tectonic units. *Swiss J Geosci* 101:139–183
- Schoonen MAA (2004) Mechanisms of sedimentary pyrite formation. *Geol Soc Am Spec Pap* 379:117–134
- Šiftar D (1988) The chemical characteristics of barite from some Bosnian deposits. *Rudarsko-metalurški zbornik* 35:75–89
- Stoll B, Jochum KP, Herwig K, Amini M, Flanz M et al (2008) An automated iridium-strip heater for LA-ICP-MS bulk analysis of geological samples. *Geostand Geoanal Res* 32:5–26
- Strashimirov S, Popov P (2000) Geology and metallogeny of the Panagyurishte ore region (Srednogorie zone, Bulgaria). In: Strashimirov S, Popov P (eds) *ABCD-GEODE 2000 workshop, guide to excursions (A and C)*. Borovets, Bulgaria, p 50
- Trubelja F, Burgath K-P, Marchig V (2004) Triassic magmatism in the area of the Central Dinarides (Bosnia and Herzegovina): geochemical resolving of tectonic setting. *Geol Croat* 57:159–170
- Vracar R, Saljić L, Sokic M, Matkovic V, Radosavljevic S (2003) Chemical-technological processing of the complex barite-sulphide ore. *Scand J Metall* 32:289–295
- Wilkin RT, Barnes HL (1997) Formation processes of framboidal pyrite. *Geochim Cosmochim Acta* 61:323–339
- Wilkin RT, Barnes HL, Brantley SL (1996) The size distribution of framboidal pyrite in modern sediments: an indicator of redox conditions. *Geochim Cosmochim Acta* 60:3897–3912
- Winchester JA, Floyd PA (1977) Geochemical discrimination of different magma series and their differentiation products using immobile elements. *Chem Geol* 20:325–343